0.1 Complete Optimization Problem

The complete constrained optimization problem is given by:

$$\begin{aligned} \underset{\tau}{\text{maximize}} \quad & g(\tau) = \alpha + \beta\tau - \gamma\tau^2 \\ \text{subject to} \quad & \tau \geq \tau_{min} \\ & \tau \leq \tau_{max} \\ & \tau \in (0, 1) \\ & \alpha > 0,\ \beta > 0,\ \gamma > 0 \end{aligned} \tag{1}$$

0.2 Optimization Method

The Karush-Kuhn-Tucker(KKT) Conditions

The following are step by step procedures of applying the model to the KKT optimization conditions:

Step 1: Re-write Constraints in Standard Form

For the KKT conditions to be applied, every constraint must be written in the form $g_i(\tau) \leq 0$. We re-write the two constraints of problem (1) as follows:

$$g_1(\tau) = \tau_{min} - \tau \leq 0 \tag{2}$$

$$g_2(\tau) = \tau - \tau_{max} \leq 0 \tag{3}$$

Step 2: Construct the Lagrangian

Since we are maximizing, the Lagrangian is formed by:

$$\mathcal{L}(\tau, \mu_1, \mu_2) = g(\tau) - \mu_1 g_1(\tau) - \mu_2 g_2(\tau) \tag{4}$$

Substituting $g(\tau)$, $g_1(\tau)$, and $g_2(\tau)$, into the Lagrangian we have:

$$\mathcal{L}(\tau, \mu_1, \mu_2) = \alpha + \beta\tau - \gamma\tau^2 - \mu_1(\tau_{min} - \tau) - \mu_2(\tau - \tau_{max}) \tag{5}$$

where $\mu_1 \geq 0$ and $\mu_2 \geq 0$ are the KKT multipliers for constraints g_1 and g_2 respectively.

Step 3: Apply the Four KKT Conditions

For τ^* to be the optimal solution of problem (1), the following four conditions must all be satisfied simultaneously.

Stationarity. We differentiate the Lagrangian with respect to τ and set it equal to zero:

$$\frac{\partial \mathcal{L}}{\partial \tau} = \beta - 2\gamma\tau + \mu_1 - \mu_2 = 0 \tag{6}$$

This equation links the optimal tax rate τ^* directly to the two multipliers μ_1 and μ_2.

Primal Feasibility. The optimal solution must satisfy all the constraints:

$$\tau_{min} - \tau \leq 0 \quad \iff \quad \tau \geq \tau_{min} \tag{7}$$

$$\tau - \tau_{max} \leq 0 \quad \iff \quad \tau \leq \tau_{max} \tag{8}$$

Dual Feasibility. The KKT multipliers must both be non-negative:

$$\mu_1 \geq 0, \qquad \mu_2 \geq 0 \tag{9}$$

Complementary Slackness. For each constraint, either the constraint is active or its multiplier is zero:

$$\mu_1(\tau_{min} - \tau) = 0 \tag{10}$$

$$\mu_2(\tau - \tau_{max}) = 0 \tag{11}$$

Step 4: For Case Analysis

From the complementary slackness conditions (10) and (11), there are three(3) important cases to consider:

Case 1: Interior Solution ($\mu_1 = 0$, $\mu_2 = 0$, both constraints inactive)

If neither constraint is binding, both multipliers are zero. Substituting $\mu_1 = \mu_2 = 0$ into the stationarity condition (6):

$$\beta - 2\gamma\tau + 0 - 0 = 0 \tag{12}$$

$$\beta - 2\gamma\tau = 0 \tag{13}$$

It implies:

$$\tau^* = \frac{\beta}{2\gamma} \tag{14}$$

Optimal Tax Rate for Case 1:

$$\tau^* = \frac{\beta}{2\gamma}$$

This is the tax rate at which the marginal benefit of taxation (captured by β) exactly equals the marginal cost of taxation (captured by $2\gamma\tau^*$). This solution is valid when $\tau_{min} < \tau^* < \tau_{max}$.

We verify whether it is a maximum by checking the second derivative of $g(\tau)$, so it implies:

$$g''(\tau) = -2\gamma < 0 \quad \text{since } \gamma > 0 \quad \checkmark$$

The negative second derivative confirms that τ^* is indeed a maximum.

Case 2: Lower Bound Active ($\tau = \tau_{min}$, $\mu_2 = 0$)

If the unconstrained optimum lies below τ_{min}, the lower bound constraint becomes binding, meaning $\tau^* = \tau_{min}$. Substituting $\tau = \tau_{min}$ and $\mu_2 = 0$ into the stationarity condition (6):

$$\beta - 2\gamma\tau_{min} + \mu_1 - 0 = 0 \tag{15}$$

$$\mu_1 = 2\gamma\tau_{min} - \beta \tag{16}$$

For this case to be valid, dual feasibility requires $\mu_1 \geq 0$:

$$2\gamma\tau_{min} - \beta \geq 0 \implies \frac{\beta}{2\gamma} \leq \tau_{min}$$

This means the unconstrained optimum falls below the minimum allowed tax rate. The lower bound constraint is therefore active, and the constrained optimum is $\tau^* = \tau_{min}$.

Case 3: Upper Bound Active ($\tau = \tau_{max}$, $\mu_1 = 0$)

If the unconstrained optimum exceeds τ_{max}, the upper bound constraint becomes binding, meaning $\tau^* = \tau_{max}$. Substituting $\tau = \tau_{max}$ and $\mu_1 = 0$ into the stationarity condition (6):

$$\beta - 2\gamma\tau_{max} + 0 - \mu_2 = 0 \tag{17}$$

$$\mu_2 = \beta - 2\gamma\tau_{max} \tag{18}$$

For this case to be valid, dual feasibility requires $\mu_2 \geq 0$:

$$\beta - 2\gamma\tau_{max} \geq 0 \implies \frac{\beta}{2\gamma} \geq \tau_{max}$$

This means the unconstrained optimum exceeds the maximum allowed tax rate. The upper bound constraint is therefore active, and the constrained optimum is $\tau^* = \tau_{max}$.

Summary of All Three Cases

$$\tau^* = \begin{cases} \tau_{min} & \text{if } \dfrac{\beta}{2\gamma} \leq \tau_{min} \\ \dfrac{\beta}{2\gamma} & \text{if } \tau_{min} < \dfrac{\beta}{2\gamma} < \tau_{max} \\ \tau_{max} & \text{if } \dfrac{\beta}{2\gamma} \geq \tau_{max} \end{cases} \tag{19}$$

Sufficiency of the KKT Conditions

Step 5: Sufficiency Verification

Having identified the candidate optimal solution τ^* from the KKT conditions, we now confirm that it is indeed a global maximum and not just any stationary point.

We check the second derivative of the objective function $g(\tau) = \alpha + \beta\tau - \gamma\tau^2$:

$$g''(\tau) = -2\gamma$$

Since $\gamma > 0$ by assumption, we have:

$$g''(\tau) = -2\gamma < 0 \quad \text{for all } \tau$$

This means $g(\tau)$ is strictly concave. For a strictly concave objective function with convex constraints, the KKT conditions are both necessary and sufficient for a global maximum. Therefore, the solution τ^* obtained from the KKT conditions is guaranteed to be a unique global maximum of the optimization problem (1).